

LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE
Final Examination Spring 2025

Class: BSCS

Subject: Computer Science

Semester: IV

Term: Final

Paper Title: Operating Systems

Max Marks: 50

Time Allowed: 3 hours

- Q:1** (a) Consider the following scenario: two processes, P1 and P2, need to access a critical section of code. Consider the following synchronization construct used by the processes: Here, wants1 and wants2 are shared variables, which are initialized to false. **[5+5=10]**

<pre>/* P1 */ while (true) { wants1 = true; while (wants2 == true): /* Critical Section */ wants1=false; } /* Remainder section */ }</pre>	<pre>/* P2 */ while (true) { wants2 = true; while (wants1==true); /* Critical Section */ wants2 = false; } /* Remainder section */ }</pre>
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Either the following statements are true or false about the above construct?

It does not ensure mutual exclusion.	
It does not ensure bounded waiting.	
It requires that processes enter the critical section in strict alternation.	
It does not prevent deadlocks, but ensures mutual exclusion.	

- (b) Explain the Security violation categories.

- Q:2** Consider the following scenarios and answer the questions:

[5+5=10]

- (a) Resource Allocation Graphs can be used to detect deadlocks from a system. Evaluate the following system:

Processes: P1, P2, P3

Resources: R1, R2, R3

Edges: P1 is waiting for a resource R1

P2 is waiting for a resource R2

P3 is waiting for a resource R3

R1 is allocated to P2

R2 is allocated to P3

R3 is allocated to P1

Is there a deadlock in this system? Justify your answer

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P.T.O
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(b) A single processor system has three resource types X, Y and Z, which are shared by three processes. There are 5 units of each resource type. Consider the following scenario, where the column alloc denotes the number of units of each resource type allocated to each process, and the column request denotes the number of units of each resource type requested by a process in order to complete execution. Write the sequence of process execution?

	Alloc			Request		
	X	Y	Z	X	Y	Z
P0	1	2	1	1	0	3
P1	2	0	1	0	1	2
P2	2	2	1	1	2	0
	0	1	2			

Q:3 Consider a disk with cylinders numbered from 0 to 199. The disk queue contains the following set of requests in the order they arrive: [10]

Disk Queue: 180, 40, 20, 75, 90, 150, 110, 130
The initial position of the disk head is at cylinder 120, and the head is currently moving toward higher-numbered cylinders.

- Determine the order in which the requests will be serviced using the following disk scheduling algorithms:
 - SCAN (Elevator Algorithm)
 - C-SCAN (Circular SCAN)
 - C-LOOK (Circular LOOK)
- For each algorithm, show the head movement and calculate the total head movements in terms of cylinders.

Q4: (a) Compare the Algorithms

Page Reference String: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7

Number of Page Frames: 4

Compute the number of page faults for the following algorithms: FIFO, LRU and Optimal and elaborate which algorithm performs best for this reference string? [6]

(b) Define how Direct Memory Access helps the hardware to access main memory. Discuss how?

